Lesson plan · for the teacher

Tidy sheets a computer can use

Data engineering · Level 2 · Lesson 1

Time
40 min

Materials
A messy paper mark sheet you made up (one per group); Rulers and squared paper; Optional: a computer or phone with a spreadsheet app and Python

Learning goals

- Lay out a sheet a computer can use.
- Keep the raw data apart from the summary.
- See how a program reads a tidy CSV file.

Activity: From messy to tidy

1. Warm-up (5 min): Show a messy mark sheet on the board, with merged class headings, a Total row and “14 / 17” in one cell. Ask pupils what a computer would get wrong.
2. Teach (8 min): Go through the rules: one header row, one row per pupil, one value per cell, no totals inside. Show a raw sheet and a summary sheet side by side.
3. Redraw (15 min): Each group gets the messy sheet and redraws it as a tidy table on squared paper, then works out the class average from it.
4. Compare (7 min): Groups write their averages on the board. Where they differ, find the cause together: a Total row counted, or a mark lost in a merged cell.
5. Wrap-up (5 min): If you have a device, show the eight lines of Python that read the tidy CSV and print the average. Then give the exit ticket.

Use made-up names on the messy sheet. Make sure the merged cells and the Total row really change the average, so the groups’ answers differ.

Discussion

- Why should nobody type over the raw sheet?
- What did a Total row do to our average?
- Why does Python need int() before adding marks?

Exit ticket

- Write the three rules of a tidy sheet, and one thing that belongs on the summary sheet.

Try it at school tomorrow
Give groups a messy paper mark sheet you made up, with merged cells, totals and two marks in a cell. They redraw it as a tidy table, then compare: whose table gives the right class average?

Practice questions · for you

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1 **Python reads every value in a CSV file as text. What must you do before adding up the marks?**

◯ Turn each one into a number with int() ◯ Delete the header row ◯ Sort the rows first

2 **When you find a mistake, it's best to type over the raw sheet, so there's only one copy of the data.**

◯ True ◯ False

3 **A mark read from a CSV file is the text "14". What does this program print?**

```
mark = "14"
print(mark + mark)
print(int(mark) + int(mark))
```

◯ 28, then 28 ◯ 28, then 1414 ◯ 1414, then 28

4 **csv is already imported (import csv). Put the lines in order and indent them: the program prints the class's average maths mark from marks.csv, leaving out pupils whose mark is blank.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `marks.append(int(row["maths"]))`

☐ `marks.append(row["maths"])`

☐ `for row in csv.DictReader(open("marks.csv")):`

☐ `print(sum(marks) / len(marks))`

☐ `marks = []`

☐ `if row["maths"] != "":`

5 **Match each spreadsheet tool to the job it does on a tidy sheet.**

AVERAGE	• • • •	Show only some rows, such as one class
COUNTIF	• • • •	Put the rows in order, such as highest mark first
Filter	• • • •	The mean of a range of marks
Sort	• • • •	How many cells match a rule, such as "6B"

✎ **Which sheet at your school is typed for people, not computers? What is the first rule you would fix?**

Check your answers

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1 **Turn each one into a number with int()**

Everything in a CSV file arrives as text.

2 **False**

Keep the raw sheet exactly as collected.

3 **1414, then 28**

Joining two pieces of text with + puts them side by side: "14" + "14" is "1414".

4 **Top to bottom: 4, X, 2, 5, 1, 3**

```
marks = []
for row in csv.DictReader(open("marks.csv")):
    if row["maths"] != "":
        marks.append(int(row["maths"]))
print(sum(marks) / len(marks))
```

The empty list comes first, so the loop has somewhere to put the marks.

5 **AVERAGE — The mean of a range of marks**

COUNTIF — How many cells match a rule, such as "6B"

Filter — Show only some rows, such as one class

Sort — Put the rows in order, such as highest mark first

Formulas such as AVERAGE and COUNTIF belong on the summary sheet; filters and sorting work on the tidy data itself.